

AnonDHT in the Boss Fight Regime: Compiling a MaxL Budget into Lookup Knobs

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Abstract

Anonymous DHT lookups are a canonical metadata problem: even if payloads are encrypted, routing structure, retries, and selective failures can identify the queried key or destination. Building on the Boss Fight accounting lens (Paper A), we give a practical *dial sheet* for anonymous DHTs: how to translate a maximal-leakage target into concrete design knobs such as dummy-query width, path splitting, timeout/abort semantics, and schedule equalization. We keep the paper intentionally tight by importing padding and stop-time machinery from earlier series papers. Instantiating the sheet on an IPFS-style Kademlia slice illustrates the central operational punchline: under modest corruption and a realistic repetition window (e.g., $Q \approx 10^3$ lookups), dummy cover must approach $w/K \rightarrow 1$, so dummy-only designs economically collapse unless observation probability is driven extremely small or queries are rate-limited.

Evidence addendum. A companion addendum collects supporting evidence tables (pattern map, contact-sweep tables, and additional knob sweeps) that are useful for auditing but kept separate to keep this dial sheet tight.[58]

Scope. This paper is a *dial sheet*: it compiles a Boss-Fight (MaxL/PML) budget into implementable Anonymous-DHT lookup knobs (dummies, path-splitting, stop-times, retries, and schedule equalization).

Imports. We import the Boss-Fight accounting lens (Bossfight A), transcript tiers (Synthesis 3), and endpoint sizing rules (Synthesis 5 / Anonymity A). Padding and stop-time normal forms are treated as owned by the published wiki posts; certificate packaging is treated as owned by ABOM/OINL, release receipts, and the receipt line-item schema.[44, 43, 45, 56]

Exports. A concrete knob taxonomy and conservative parameter-mapping sufficient to populate ABOM/OINL and to support change-control recertification when knobs drift.

Artifact policy. Source-only; plots, code, and deployment traces are available on request.[44]

Build note. Default build includes the core quantitative tables (pure L^AT_EX evidence).

1 Scope and relationship to Paper A

Paper A [30] defines MaxL/PML budgets and shows why repeated use is the correct regime. This paper specializes those ideas to *Anonymous DHT lookups*: a conservative mapping from a Boss Fight target to implementable parameters.

Table 1: Concrete Kademlia parameters in the IPFS DHT (Amino swarm) [17]. These defaults make the “Boss Fight” repetition regime unavoidable unless the lookup transcript is aggressively equalized or cryptographically hidden.

Item	IPFS Kademlia DHT spec value
Identifier size	256-bit Kademlia identifiers for peers and content routing
Bucket size	$k = 20$ closest peers returned per query (routing table buckets)
Lookup termination	resilience parameter $\beta = 3$ closest reachable peers queried
Lookup concurrency	maximum in-flight queries $\alpha = 10$
Provider routing	provider lookup/advertise routed to the k closest peers

2 Background and pointers

Structured overlays (DHTs) such as Chord and Kademlia provide efficient key-based routing [3, 4]. For anonymous lookups, the dominant leakage channels are not payload content but *transcripts*: which peers are contacted, in what order, with what retries/timeouts, and under what failure patterns. We use the observation-tier vocabulary and exposure normal forms from the companion trace-interface note [31] to keep this dial sheet non-redundant. Classic attack analyses show that passive range estimation and active selective failures can collapse anonymity in DHT-based relay selection unless the system budgets retries and equalizes visible behavior [9]. Octopus is a representative “secure+anonymous lookup” design that instantiates many of the knobs we compile here (path splitting, dummies, and stopping rules) [10].

This paper is intentionally a *dial sheet*: it maps a MaxL target to knobs. For broader background on transcript witnesses and how to size them under repeated use, see the series odds-inflation and profiling papers [35, 33].

Pointer. A companion addendum includes a compact knob-oriented map of classic AnonDHT designs (NISAN/Torsk/Octopus/IRIS/query-obfuscation) to orient where each dial appears. [58]

Information-theoretic analyses of structured overlays. Beyond protocol-specific attacks, there is a line of work modeling privacy loss in DHTs via information-theoretic quantities derived from routing state and geometry (e.g., compromised routing tables) [29]. Our contribution differs in that we size *repeated-use identification risk* using MaxL/PML budgets (Paper A), and then compile those budgets into concrete lookup knobs.

Concrete deployment anchor (brief). As a sanity check on scale, widely deployed Kademlia-style systems (e.g., IPFS/libp2p) typically expose on the order of tens of peer contacts per lookup, with retry/failure tails. Table 1 captures representative public defaults; the long build adds a short mini-spec slice and additional deployment pointers.

3 Threat model and observation probability

We treat the *lookup target* C (key, destination, or shard) as the secret and the adversary’s extracted transcript feature as Y (Paper A). In the Boss Fight, the adversary aggregates many lookups and may induce retries, delays, or selective failures.

Declaration (non-redundancy). We use the threat/window checklist of the companion note (Synthesis 4) [32] and the tier/exposure vocabulary of the trace-interface note [31]. For this dial sheet we take *secret* = lookup target C (key/destination/shard) and treat the adversary’s extracted transcript feature as Y (Paper A). The *unit* is a lookup (or a lookup attempt, depending on the declared projection) and the *window* is an operational query budget Q per time window with an explicit reset rule.

Passive upgrades vs Boss Fight (interpretation). Many deployed “reader privacy” proposals for Kademlia-derived systems primarily target passive observers; in that regime, treat this dial sheet as a conservative stress test sized for repetition and induced retries [46].

What does p_{obs} mean? Observation probability is tier-relative and must match the declared projection: at Tier 2 (ordered contact/failure traces) p_{obs} is close to 1 for compromised/queried peers, while at lower tiers or with cryptographic reader-privacy layers it can be smaller. If a deployment publishes multiple witness tiers (contact-set vs timing vs congestion), receipts should publish a tiered observation vector (one (p_i, b_i) pair per tier) and combine tiers using the conservative rule in Synthesis 15.[57] We avoid re-tutorializing tiering here; we only require that the deployment declares which projection is being bounded and audits that the logging/monitoring pipeline matches it [31, 32].

What is the candidate universe K ? The relevant $K \triangleq |\mathcal{C}|$ is the number of *plausible* targets under declared public side information in the declared window; see the threat/window checklist [32] and Paper C’s certificate fields.

Many AnonDHT proposals contact *committees* (shards) of size s sampled from N nodes, with an adversary corrupting a ρ fraction. A pessimistic sufficient condition for “observing” a contacted committee is that it contains at least one corrupted node. Then the per-contact observation probability is

$$p_{\text{obs}} = 1 - \frac{\binom{(1-\rho)N}{s}}{\binom{N}{s}} \approx 1 - (1 - \rho)^s. \quad (1)$$

For example, with $\rho = 0.2$ and $s = 10$, $p_{\text{obs}} \approx 0.893$.

In Kademlia-style iterative lookups, the number of distinct queried peers m can be substantial (often $\mathcal{O}(10)$ and sometimes much larger under failures/retries). We treat m as a knob-dependent scale parameter; representative measurements put m in the ~ 10 – 20 range for most lookups with a long tail under failures/retries [17, 28, 19, 25].

Caution (independence proxy). The proxy $p_{\text{obs}} = 1 - (1 - \rho)^m$ treats each contacted peer as independently adversarial. Real overlays exhibit correlations (routing-table poisoning, selective targetting, and churn), so we use this proxy only as a conservative *scale anchor* rather than a claim of tightness.

Lemma 1 (Contacts amplify observation). *If each queried peer is independently adversarial with probability ρ (or, more generally, the adversary observes a query whenever it hits an adversarial peer), then the probability that a lookup has any observed contact satisfies*

$$p_{\text{obs}}^{(\text{lookup})} \geq 1 - (1 - \rho)^m,$$

where m is the number of distinct queried peers.

Proof sketch. The lookup is unobserved only if *all* m queried peers are non-adversarial. □

Numerical note. In realistic repetition windows, once p_{obs} is not tiny and the per-lookup budget b is forced into the 10^{-2} – 10^{-3} bit range, the required dummy fraction approaches 1. [58]

4 A minimal AnonDHT slice with a closed-form MaxL bound

To keep the paper tight, we isolate a single lookup pattern that appears (explicitly or implicitly) across many AnonDHT designs: *forced-destination with decoys* under *partial visibility*.

Slice. Let the secret lookup target be $C \in \mathcal{C}$ with $|\mathcal{C}| = K$ possibilities. A lookup instance selects a *candidate set* $S \subseteq \mathcal{C}$ of size $w+1$ that contains the true target C and w decoys (e.g., drawn uniformly from $\mathcal{C} \setminus \{C\}$). The mechanism then runs the protocol on each element of S (possibly interleaving or batching queries). The adversary’s *observation event* is modeled as: with probability p_{obs} the transcript reveals S (e.g., via corrupted committee members, prefix disclosure, or selective-failure side channels), and with probability $1 - p_{\text{obs}}$ the transcript reveals no target-dependent identifier (an erasure symbol $\perp$). This yields an explicit channel $C \mapsto Y \in \{\perp\} \cup \binom{\mathcal{C}}{w+1}$.

Theorem 1 (Forced-destination+ w -dummies under erasure). *In the slice above,*

$$\mathcal{L}(C \rightarrow Y) = \log_2\left((1 - p_{\text{obs}}) + p_{\text{obs}} \cdot \frac{K}{w+1}\right) = \log_2\left(1 + p_{\text{obs}}\left(\frac{K}{w+1} - 1\right)\right).$$

Derivation. This is an immediate corollary of Paper A’s forced-destination-with-dummies MaxL identity and its erasure/partial-visibility rule, restated here for convenience.

Corollary 1 (Dummy width required for a per-lookup budget). *If we require $\mathcal{L}(C \rightarrow Y) \leq b$ for one lookup in the slice, then necessarily*

$$w+1 \geq \frac{p_{\text{obs}}K}{2^b - (1 - p_{\text{obs}})} \quad (\text{when } 2^b > (1 - p_{\text{obs}})).$$

In particular, if $b = 0$ then $w \geq K-1$ (full cover), and when p_{obs} is close to 1, w must be a large fraction of K even for b near 1 bit.

5 Budgeting a lookup under repetition

Let a single lookup produce step observations $Y_1, \dots, Y_L$ (routing hops, committee contacts, abort/timeout events). Paper A’s accountant gives

$$\mathcal{L}(C \rightarrow Y_{1:L}) \leq \sum_{\ell=1}^L \mathcal{L}(C \rightarrow Y_\ell \mid Y_{<\ell}) \triangleq \Lambda_{\text{lookup}}.$$

Over Q lookups in a window, the conservative Boss Fight budget is $\Lambda_{\text{window}} \leq Q\Lambda_{\text{lookup}}$. Thus even “small” per-lookup leakage becomes large at scale.

5.1 Worked sizing example: why repetition dominates

The uniform-guess bound from Paper A (and its conservative “Boss Fight firewall” repetition-cap corollary) implies that keeping an analyst’s success probability below δ over a window of Q lookups requires a *total* budget $\Lambda \leq \log_2(\delta K)$ for K plausible targets. Thus the per-lookup average budget is $b = \Lambda/Q$, which quickly becomes $\ll 1$ bit. In the minimal slice (Section 4), such tiny b forces the dummy width w to be nearly the entire candidate universe unless p_{obs} is extremely small.

Table 2: Boss Fight budgeting forces per-lookup leakage to be tiny under repetition. If a window allows total leakage B bits over Q lookups, the per-lookup target is $b = B/Q$.

Q lookups/window	$B=2$ bits	$B=5$ bits	$B=10$ bits
10	0.2000	0.5000	1.0000
100	0.0200	0.0500	0.1000
1,000	0.0020	0.0050	0.0100
10,000	0.0002	0.0005	0.0010

Table 3: Worked Boss Fight sizing: $K = 2^{20}$ plausible targets, total guess cap $\delta = 10^{-3}$, and committee observation probability $p_{\text{obs}} = 0.5$. Total allowable budget is $\Lambda = \log_2(\delta K) \approx 10.03$ bits; repetition forces a tiny per-lookup budget $b = \Lambda/Q$, which in turn forces near-full dummy cover by Corollary 1.

Q lookups	per-lookup $b = \Lambda/Q$ (bits)	min. w	w/K
10	1.0034	348,422	0.332
100	0.1003	916,543	0.874
1000	0.0100	1,034,140	0.986
10000	0.0010	1,047,118	0.999

5.2 End-to-end window example (IPFS-style Kademlia)

For a concrete instantiation, we plug a widely deployed IPFS Kademlia parameterization into the minimal slice. The IPFS spec uses $k=20$, concurrency $\alpha=10$, and termination threshold $\beta=3$ [17]. Under a high-concurrency stress test, an IPFS DHT node observed that 99% of lookups complete within 18 hops, with a rare long tail [28]. Treating each hop as a distinct contacted peer gives a conservative contact count $m = 18$. Combining this with a simple corruption-to-observation proxy $p_{\text{obs}} = 1 - (1 - \rho)^m$ yields the end-to-end sizing in Table 5.

6 Dial sheet: knobs and what they buy

Table 6 is the practical summary.

7 Why dummy cover alone usually fails

A common pattern is “forced destination + dummies”: contact the true shard plus w dummy shards. Paper A gives the erasure-adjusted bound

$$\mathcal{L} = \log_2\left((1 - p_{\text{obs}}) + p_{\text{obs}} \cdot K/(w + 1)\right),$$

where K is the number of plausible destinations and p_{obs} is the probability the adversary *sees* the contact. Solving $\mathcal{L} \leq b$ yields a required dummy width; rather than restate it, we refer to Corollary 1 (derived earlier for the same slice).

Table 4: Visibility sensitivity at fixed repetition: $K = 2^{20}$, $Q = 1000$, total guess cap $\delta = 10^{-3}$, so $b = \log_2(\delta K)/Q \approx 0.010$ bits per lookup. Even when p_{obs} is modest, the dummy set must cover most candidates.

p_{obs}	min. w (for $b = \Lambda/Q$)	w/K
0.100	980,165	0.935
0.200	1,013,217	0.966
0.500	1,034,140	0.986
0.893	1,040,444	0.992
0.950	1,040,928	0.993

Window lookups Q	per-lookup budget $b = \Lambda/Q$ (bits)	required w/K ($\rho=5\%$)	required w/K ($\rho=10\%$)
100	0.1003	0.893	0.922
1,000	0.0100	0.989	0.992
10,000	0.0010	0.999	0.999

Table 5: End-to-end Boss Fight sizing for an IPFS-style Kademlia lookup slice. Assume a candidate set of size $K = 2^{20}$ (e.g., a 20-bit prefix bucket after prior information), and a window success target $\delta = 10^{-3}$, so $\Lambda = \log_2(\delta K) \approx 10.03$ bits total. Assume the lookup contacts $m = 18$ peers (a conservative 99% hop/contact bound reported in an experiment with the libp2p/IPFS implementation [28]), and each contacted peer is independently corrupted with probability ρ , giving $p_{\text{obs}} = 1 - (1 - \rho)^m$ (here $p_{\text{obs}} \approx 0.60$ for $\rho = 5\%$ and 0.85 for $\rho = 10\%$). The table reports the dummy width fraction w/K required in the minimal slice of Section 4 to meet $\text{MaxL} \leq b$. Even moderate repetition forces $w/K \rightarrow 1$.

Approximation. When b is tiny, $2^b - 1 \approx (\ln 2) b$. If p_{obs} is not itself tiny, Corollary 1 implies the scaling

$$w \gtrsim \frac{p_{\text{obs}} K}{p_{\text{obs}} + (\ln 2) b} - 1 \approx K \left(1 - \frac{(\ln 2) b}{p_{\text{obs}}} \right),$$

i.e., w must cover essentially the entire candidate universe.

7.1 When can cover work?

Corollary 1 has a sobering consequence: for large K , the required *fraction* w/K depends mainly on (p_{obs}, b) , not on K itself. Thus, “just shrink the universe” is not enough if the Boss Fight drives b into the 10^{-2} – 10^{-3} bit range. Cover becomes practical only in regimes where (i) rate limiting (or coarse bucketing) keeps b moderately large, and/or (ii) the implementation makes p_{obs} extremely small by shielding contacts. Table 8 makes this explicit.

In the large- K regime, ignoring the $\lceil \cdot \rceil$ rounding in Corollary 1, one obtains the convenient approximation

$$\frac{w}{K} \approx \frac{p_{\text{obs}}}{2^b - 1 + p_{\text{obs}}}. \quad (2)$$

Implication for AnonDHT design. If your use case requires many lookups per window, you should plan on mechanisms that *reduce observation* or *hide the queried key* rather than only widening dummy sets. In IPFS, this motivates both (a) delegated routing reader-privacy upgrades (en-

crypting content-routing requests in the HTTP delegated routing API) [52, 53] and (b) PIR-style approaches such as Peer2PIR [49].

8 Takeaways

- **Dial the Boss Fight with two numbers:** observation probability p_{obs} and per-lookup MaxL budget b essentially determine the dummy fraction via Eq. (2) and Corollary 1.
- **Shrinking K is not enough:** in the large- K regime, w/K depends mainly on (p_{obs}, b) , so “smaller universes” do not rescue low-latency anonymity once b collapses into the 10^{-2} – 10^{-3} bit range.
- **Prefer hiding visibility over widening cover:** to keep overhead practical you must keep b moderately large (rate-limit / bucket) and/or drive p_{obs} extremely small (shield contacts); otherwise cryptographic query privacy and reader-private routing (e.g., Peer2PIR; delegated-routing reader privacy) dominate cover work.
- **How to cite / reuse:** use BossFight A for budget definitions and MaxL $\leftrightarrow$ PML equivalences; cite Anonymity B for the profiling contract and Operational A for retry discipline.

9 Imports from earlier series papers

For long-horizon profiling and committee-trace equalization: [33]. For stateful anonymity, committee-contact privacy, and closed-view auditing (to avoid re-deriving state games here): [36, 37, 38]. For concrete padding constructions and dual certificates: [34, 39, 40, 42]. For lower bounds explaining why low-latency anonymity is hard at scale: [41].

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Table 6: AnonDHT dial sheet (Boss Fight view). Each knob reduces one class of destination-indexed witness.

Knob	What it controls in the MaxL/PML budget
Dummy width (w)	Hides a forced destination among w dummies. Under full visibility, MaxL is $\log_2(K/(w+1))$ for K plausible destinations (Paper A). Under high p_{obs} , erasure helps little.
Path splitting / query splitting	Spreads one high-leak contact into multiple lower-leak contacts, but increases the number of observed events; must be budgeted.
Query obfuscation (Kademlia)	Obfuscates the queried identifier (e.g., double hashing, PSI-based query hiding, and prefix fetching), trading extra contacts/bytes for per-contact ambiguity; the added contacts must still be budgeted (and can increase p_{obs}) [46].
Query transport encryption	Reduces observation by hiding queried identifiers from passive network observers or routing services (e.g., Discv5 encrypts discovery traffic [20]; delegated routing reader-privacy proposals encrypt content-routing requests [52, 53]). Does not hide the key from the contacted peer itself; colluding peers still induce high p_{obs} .
Caching / prefetch	Reduces the number of fresh lookups Q per window (and hence the linear Boss Fight accumulation), but can introduce new side channels (cache hits/misses) if not equalized.
Schedule equalization	Forces timing witnesses into coarse bins; imported from [34].
Timeout/abort policy	Critical: aborts and retries are destination-indexed witnesses under selective failure and probing; see [1, 12]. Aborts are secret-indexed injections. In deployed DHTs this channel is actively exploited (e.g., Kubo early-termination triggers under Sybil placement) [27]. Must be bounded and padded (Paper A; [42, 40]).
Retry cap / stop-time padding	Bounds the number of “chances” the analyst gets; imported from [42].
Committee sampling / reshuffling	Controls witness overlap across repeated lookups; higher-order KICS constraints may be required (Paper A).
Private retrieval / OT in routing	Reduces the effective destination universe (or hides the key inside a shard) by cryptographic private-query primitives (authenticated PIR, OT-based query privacy) [15, 50].

Table 7: When the effective universe K is small (e.g., per-hop bucket or shortlist) *and* observation probability is low, dummy cover can be a viable knob. We show the smallest w satisfying Corollary 1 with $p_{\text{obs}} = 0.2$.

K (candidates)	$b=0.5$ bits	$b=1$ bit
2^5	10	5
2^6	20	10
2^7	41	21
2^8	83	42

per-lookup budget b (bits)	$p_{\text{obs}}=0.01$	0.05	0.2	0.5
1.00	0.010	0.048	0.167	0.333
0.50	0.024	0.108	0.326	0.547
0.10	0.122	0.411	0.736	0.874
0.01	0.590	0.878	0.966	0.986

Table 8: Required dummy *fraction* w/K in the minimal slice of Section 4, ignoring the negligible $1/K$ rounding, as a function of observation probability p_{obs} and per-lookup MaxL budget b . For large K , the fraction is approximately $w/K \approx p_{\text{obs}}/(2^b - 1 + p_{\text{obs}})$ (Eq. (2)). Thus shrinking K alone does *not* fix the Boss Fight: you must keep b from becoming tiny (rate-limit) and/or drive p_{obs} extremely small (shield contacts).